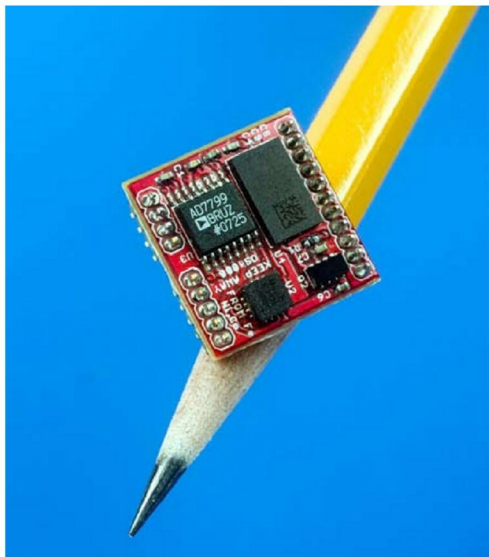


# SMT COMPONENTS THE SMALL WONDER

**Small is big business not just in automaking but in electronics too. And, the flag bearers of this change are the surface-mount components**



■ K.T.P. RADHIKA

**W**hile using a smart phone that performs all the scintillating digital magic earlier done by a computer,

have you ever fancied what triggers the wonder? Or what helped these devices shrink their size while enduring power and performance?

Well, the wizard here is called surface-mount components—said to be the lifeline of surface-mount technology.

Any electronic device—from LCD TVs to the latest small mobile phones—is today packed with these tiny devices. These devices help manufacturers pack more electronic components in a much smaller space while enabling more functionality.

## SMT vs through-hole technology

The evolution of surface-mount technology dates back to the 1960s. However, this technology caught the fancy of users across the globe only in the 1980s. (For more details, see April 2008 issue of EFY.) Unlike its popular counterpart through-hole technology, components in SMT are directly mounted onto a printed circuit board (PCB). In through-hole, pins are inserted into drilled holes on a PCB.

SMT components have small metallic end caps which are directly soldered onto the PCB surface. This technology reduces the package size and improves functionality to a great extent on the same board-area. SMT places components on both sides of the PCB, which helps to reduce time in assembling the components. Placing components on both sides of the board also allows much higher circuit densities.

Surface-mount devices are very small and thin, compared to through-hole. Specifically speaking, these are one-quarter to one-tenth the size and weight, and one-half to one-quarter the cost of equivalent through-hole parts. This allows lower resistance and inductance at the connection, leading to